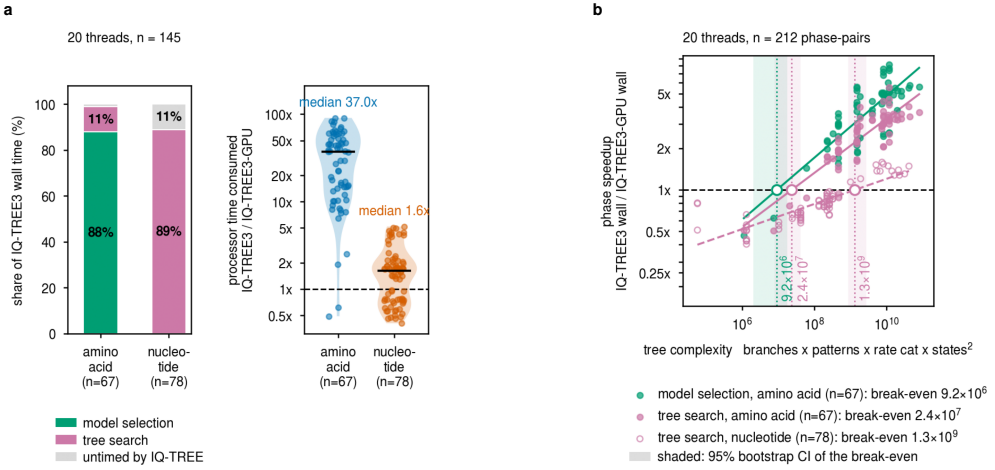
Figure 4


a
20 threads, n = 145
share of IQ-TREE3 wall time (%)
11%
88%
11%
89%
amino acid (n=67)
nucleo-tide (n=78)
processor time consumed IQ-TREE3 / IQ-TREE3-GPU
median 37.0x
median 1.6x
model selection
tree search
untimed by IQ-TREE
b
20 threads, n = 212 phase-pairs
phase speedup IQ-TREE3 wall / IQ-TREE3-GPU wall
9.2×10⁶
2.4×10⁷
1.3×10⁹
tree complexity branches x patterns x rate cat x states²
model selection, amino acid (n=67): break-even 9.2×10⁶
tree search, amino acid (n=67): break-even 2.4×10⁷
tree search, nucleotide (n=78): break-even 1.3×10⁹
shaded: 95% bootstrap CI of the break-even
c
20 threads, n = 145
amino acid (n=67)
nucleotide (n=78)
20 cores requested
cores kept busy (processor time / wall time)
IQ-TREE3
IQ-TREE3-GPU
IQ-TREE3
IQ-TREE3-GPU